Validation with the Blasius Equation

In this document, I will be discussing the Blasius Equation and (shortly) how it was used to compare with the accompanying code in the Github repo.

This document will go over the derivation of the Blasius Equation and a little bit about the validation which includes what quantities were compared and a small bit about the mesh and BCs.

Honestly, this document is more for me so that I can keep track of how I did things and is really just good for learning.

Okay, let's begin!

Contents

Blasius Equation 2

Overview 2

Derivation 2

Governing Equation 2

Order of Magnitude Analysis 2

External Flow 5

Boundary Layer Transformations 6

Zero Pressure Gradient (m=0) 9

Validation 10

Approach 10

Mesh 12

Blasius Equation

Overview

First, what is the Blasius Equation? The Blasius equation is a third-order non-linear ODE that describes the steady 2D laminar BL over a flat plate as shown in Equation (1).

$$f''' + \frac{1}{2} f f'' = 0 \tag{1}$$

$$\text{where} \quad f(0) = 0, \qquad f'(0) = 0, \qquad f'(\infty) = 1$$

It is a boundary layer approximation of the Navier-Stokes equation. It gives a similarity solution for the velocity field in a boundary layer, reducing the problem to a single dimensionless function $f(\eta)$ where η is the similarity variable. This was actually the product of Paul Blasisus for a Phd thesis (quite an impressive undertaking).

More information on this can be found online anywhere. It is one of those concepts you learn in a fundamental aerodynamics course i.e. it is not niche.

So, I will not go into the details too much because the next section, **Derivation**, will get into the details.

Derivation

Governing Equation

Okay, let's go from the start. This is a BL approximation of the N-S equations, so naturally, the N-S equations are our starting point. Let us consider the steady incompressible continuity and N-S equation shown in Equation (2) and (3), respectively.

$$\nabla \cdot u = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{2}$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial u}{\partial x^2} + \frac{\partial v}{\partial y^2} \right) \tag{3}$$

We want to perform an order of magnitude analysis to see if any of these terms might drop out in the boundary layer which will simplify the equations we need to solve.

Order of Magnitude Analysis

We can consider the following orders of magnitude for the following terms:

- $\frac{\partial}{\partial x} \sim \frac{1}{L}$ → assuming that changes in the x-direction happen over a length scale L (this is usually some characteristic length e.g. chord)
- $\frac{\partial}{\partial y} \sim \frac{1}{\delta}$ → assuming that changes in the y-direction happen over a length scale δ (this is the BL thickness)
- $\delta \ll L$ → assuming that the boundary layer is thin
- $U \sim U_\infty$ → velocity is *on the order of* U_∞
    - we are not saying that $U = U_\infty$ inside the BL but is on the order of it
    - $\frac{U}{U_\infty}$ varies between 0 and 1 in the BL so it is $O(1)$

Okay, we have an order of magnitude for all the terms in Equation (2) and (3) except for V. We can obtain an estimate for the order of magnitude of V by using the continuity equation and solving for V as below considering the diagram that follows.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \rightarrow \frac{U_\infty}{L} \sim \frac{V}{\delta} \rightarrow V \sim \delta \frac{U}{L} \tag{4}$$

It will be noted, as was in my class when I was taught this, $v \neq 0$. This is NOT a parallel flow so the v will vary throughout the BL.

Now we actually have an order of magnitude for all the terms in Equation (2) and (3). The next step is to apply those orders of magnitude to the steady incompressible 2D N-S equations and see which terms become small (and therefore able to be removed from the equation).

<u>X-Momentum Equation</u>

Recall the x-momentum equation as in Equation (5) shown below and the order of magnitudes substituted as in Equation (6).

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \tag{5}$$

$$U_\infty \frac{U_\infty}{L} + \delta \frac{U_\infty}{L}\frac{U_\infty}{\delta} \sim ? + \nu\frac{U_\infty}{L^2} + \nu\frac{U_\infty}{\delta^2} \tag{6}$$

Quick side note because I was confused:

$\frac{\partial^2 u}{\partial x^2} \sim \frac{U_\infty}{L^2}$ not $\frac{\partial^2 u}{\partial x^2} \sim \frac{U_\infty^2}{L^2}$ because we are differentiating with respect to space. Easier to see this built up. Consider $\frac{\partial u}{\partial x}$ that is clearly $\frac{\Delta u}{\Delta x} \sim \frac{U_\infty}{L}$ if we then differentiate that with respect to space again we get $\frac{\partial}{\partial x}\left(\frac{U_\infty}{L}\right) \sim \frac{U_\infty}{L^2}$.

Let us simplify and note terms that we can drop as in Equation (7) below.

$$\frac{U_\infty^2}{L} + \frac{U_\infty^2}{L} \sim ? + \nu\frac{U_\infty}{L^2} + \nu\frac{U_\infty}{\delta^2} \tag{7}$$

- Considering small or zero pressure-gradient, the "?" drops out
- Combine $\frac{U_\infty^2}{L} + \frac{U_\infty^2}{L} \sim \frac{U_\infty^2}{L}$
- On the RHS, $\nu\frac{U_\infty}{L^2}$ can drop out because $\nu\frac{U_\infty}{L^2} \ll \nu\frac{U_\infty}{\delta^2}$ because $\delta \ll L$

Hence, we are left with the following Equation (8).

$$\frac{U_\infty^2}{L} \sim \nu\frac{U_\infty}{\delta^2} \tag{8}$$

So, the only term that actually dropped out is $\nu\frac{U_\infty}{L^2}$ in the x-momentum equations. Therefore, for x-momentum, we arrive at Equation (9) following of OoM analysis.

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial y^2}\right) \tag{9}$$

Y-Momentum Equation

Recall the y-momentum equation as in Equation (10) shown below and the order of magnitudes substituted as in Equation (11).

$$u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) \tag{10}$$

$$U_\infty\frac{\delta\frac{U}{L}}{L} + \delta\frac{U_\infty}{L}\frac{\delta\frac{U}{L}}{\delta} \sim ? + \nu\frac{\delta\frac{U}{L}}{L^2} + \nu\frac{\delta\frac{U}{L}}{\delta^2} \tag{11}$$

Again, let us simplify and note terms that we can drop as in Equation (12) below.

$$\delta\frac{U_\infty^2}{L^2} + \delta\frac{U_\infty^2}{L^2} \sim ? + \nu\delta\frac{U_\infty}{L^3} + \nu\frac{U_\infty}{\delta L} \tag{12}$$

- Considering small or zero pressure-gradient, the "?" drops out
- Combine $\delta\frac{U_\infty^2}{L^2} + \delta\frac{U_\infty^2}{L^2} \sim \delta\frac{U_\infty^2}{L^2}$
- On the RHS, $\nu\delta\frac{U_\infty}{L^3}$ can drop out because both $\frac{1}{L^3}$ and δ are small making the whole term small

So, we are left with Equation (13) which is rewritten to factor out $\frac{\delta}{L}$.

$$\left(\frac{\delta}{L}\right)\left(\frac{U_\infty}{L}\right) \sim \left(\frac{\delta}{L}\right)\left(\nu\frac{U_\infty}{\delta^2}\right) \tag{13}$$

Notice that this is the **same as Equation (8) besides an additional factor of $\boldsymbol{\delta/L}$**. And, we know that the additional factor is small because the BL is small. Therefore, the x-momentum equation is >> than the y-momentum equation. So, we can disregard that y-momentum equation in considering the BL. This is a major simplification …

Equations for Approximate BL Solution

Therefore, we arrive at a simpler set of equations for our approximate BL solution as shown in Equation (14) and (15) for the continuity and x-momentum as shown. Recall that the y-momentum equation has dropped out.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{14}$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial y^2}\right) \tag{15}$$

External Flow

Recall that we are going to want to express Equation (15) as an ODE. In order to assist in the formulation, we are going to want to express the pressure gradient in terms of velocity i.e. reduce the number of unknowns. This would simplify the problem as then we would only need to consider u and v.

The way this is done is by considering what the pressure gradient is at the boundary layer edge. In the boundary layer, we know that $\frac{\partial p}{\partial y}$ is small (this can be proven through another OoM analysis that I will not be doing here). Therefore, $p_e(x) = p(x, y)$ i.e. pressure at the edge of the boundary layer is equal to that in the boundary layer.

With this, let's consider Equation (15) in the limit that $u \to U_\infty$ as $y \to edge$ where gradients of u with respect to y go to zero at the edge (where everything is at freestream). We recover Equation (16) for the edge then.

$$U_\infty\frac{\partial U_\infty}{\partial x} = -\frac{1}{\rho}\frac{dP}{dx} \tag{16}$$

What we have done is **express the pressure-gradient in terms of freestream velocity variations**. We then recover,

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = U_\infty\frac{\partial U_\infty}{\partial x} + \nu\frac{\partial^2 u}{\partial y^2} \tag{15}$$

Boundary Layer Transformations

The solution approach is to reduce the boundary-layer PDEs to an ODE by introducing a stream function and a similarity transformation. The new independent variable is the similarity variable η. The new dependent variable is the dimensionless similarity function $f(\eta)$. You solve for $f(\eta)$, then reconstruct the stream function $\psi(x, y)$.

Let's follow this by first introducing a stream function of the following form in Equation (17):

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x} \tag{16}$$

$$\psi(x, y) = g(x) f(\eta) \tag{17}$$

Where η is to be non-dimensional and we will also seek that $f(\eta)$ will.

HOLD UP… for someone looking at this the first time it is overwhelming and seems like arbitrary *magic*. How did we choose ψ and what does $g(x)$, $f(\eta)$, and η represent?

First off, we are using the stream function since that automatically satisfies the continuity equation ($\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$) while reducing the problem to one variable to solve for ψ.

Second, the basis for having a $\psi(x, y) = g(x) f(\eta)$ first comes from the observation that the boundary layer profile looks the same at every x it's just stretched vertically. The shape is, well, self-similar. In Equation (17), the terms can be interpreted as such:

- $g(x)$ → scaling function that captures how the magnitude and overall thickness of the BL grows as you move downstream in x
- $f(\eta)$ → a function that capture the shape profile of the boundary layer (naturally this should be non-dimensional since the scaling of the profile is from $g(x)$)

Third, to pick an η, dimensional analysis was considered. We want η to be dimensionless. In the context of what we want $f(\eta)$ to represent, it should really be some non-dimensional wall distance that is the input. Therefore we use $\eta = \frac{y}{g(x)/U_\infty}$. Key is that it is non-dimensional and is just y divided by a characteristic length.

At this point, $g(x)$ is unknown… we are going to determine it so that we can ultimately write everything in terms of f as in Equation (1).

Getting back to the derivation, let us rewrite Equation (15) in terms of the stream function.

$$\frac{\partial \psi}{\partial y}\frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial \psi}{\partial x}\frac{\partial^2 \psi}{\partial y^2} = U_\infty \frac{dU_\infty}{dx} + \nu \frac{\partial^3 \psi}{\partial y^3} \tag{18}$$

Write Equation (18) in terms of f, η, g, and constants (i.e. ν and U_∞). To do this we need to express the derivatives of ψ in terms of these terms. I will show a few as an example but these terms can become many terms combined with other terms and … expression swell.

$$\frac{\partial \psi}{\partial y} = g(x)\frac{\partial f}{\partial y} = U_\infty f' \tag{19}$$

$$\frac{\partial \eta}{\partial x} = \eta\left[\frac{1}{U_\infty}\frac{dU_\infty}{dx} - \frac{1}{g}\frac{dg}{dx}\right] \tag{20}$$

$$\frac{\partial^2 \psi}{\partial x \partial y} = \frac{\partial}{\partial x}\left[\frac{\partial \psi}{\partial y}\right] = U_\infty f''\frac{\partial \eta}{\partial x} + f'\frac{\partial U_\infty}{\partial x} \tag{21}$$

After all is said and done… just taking the equation we end up with from my class notes we get Equation (22)

$$f''' + \frac{g}{\nu U_\infty}\frac{dg}{dx}ff'' + \frac{g^2}{\nu U_\infty^2}\frac{dU_\infty}{dx}(1 - f'^2) = 0 \tag{22}$$

Since we are looking for a similarity solution in which f and its derivatives are non-dimensional, that means that the coefficients in purple must also be non-dimensional. Both coefficients have a dependency on x which is not something we want because f should be some universal profile based on η alone. Therefore both coefficients, which we will call α and β (as defined below) must be constant…

$$\alpha = \frac{g}{\nu U_\infty}\frac{dg}{dx} \tag{23}$$

$$\beta = \frac{g^2}{\nu U_\infty^2}\frac{dU_\infty}{dx} \tag{24}$$

Note that we can arbitrarily choose $\alpha = 1$. Why? This is because g is not a fixed term… we made it up. Therefore, we can manipulate it so that $\alpha = 1$. We could have just as easily used $\alpha = 2$. We can then focus just on β.

Both α and β contain derivatives of g and U_∞, respectively. Therefore, we look for a way to combine them. The trick is that α contains a derivative with respect to g and β contains a derivative with respect to U_∞. We then look at the derivative of $\frac{g^2}{U_\infty}$ using the quotient rule for simplifications (this is somewhat an arbitrary decision (at least to me)). This gives us:

$$\frac{d}{dx}\left(\frac{g^2}{U_\infty}\right) = \frac{1}{U_\infty}\frac{d(g^2)}{dx} - \frac{g^2}{U_\infty^2}\frac{dU_\infty}{dx} \tag{25}$$

The first term can be simplified as such.

$$\frac{1}{U_\infty}\frac{d(g^2)}{dx} = \frac{2g}{U_\infty}\frac{dg}{dx} = 2\alpha\nu \tag{26}$$

The second term can simplified as such.

$$\frac{g^2}{U_\infty^2}\frac{dU_\infty}{dx} = \beta\nu \tag{27}$$

So, the derivative of $\frac{g^2}{U_\infty}$ is a constant! If the derivative of something is a constant, that means that the thing itself is just that constant times x. Therefore,

$$\frac{g^2}{U_\infty} = (2\alpha\nu - \beta\nu)x + C \tag{28}$$

$$also\ note \rightarrow g^2 \propto U_\infty$$

Since $g(0) = 0 \rightarrow C = 0$ i.e. the boundary layer has zero thickness at $x = 0$, which is the leading edge. So, we end up with the following expression.

$$g^2 = U_\infty(2\alpha - \beta)\nu x \tag{29}$$

With this, let us solve for $U_\infty(x)$ by plugging g^2 back into β.

$$\beta = \frac{U_\infty(2\alpha - \beta)\nu x}{\nu U_\infty^2}\frac{dU_\infty}{dx} = \frac{(2\alpha - \beta)x}{U_\infty}\frac{dU_\infty}{dx} \tag{30}$$

$$\frac{1}{U_\infty}\frac{dU_\infty}{dx} = \frac{\beta}{2\alpha - \beta}\frac{1}{x} \tag{31}$$

$$\ln(U_\infty) = \frac{\beta}{2\alpha - \beta}\ln(x) + C \tag{32}$$

Therefore, finally…

$$U_\infty(x) = Cx^m \tag{33}$$

$$\text{where } m = \frac{\beta}{2\alpha - \beta}$$

Considering that we set $\alpha = 1$ arbitrarily earlier, we recover that.

$$\beta = \frac{2m}{m+1} \tag{34}$$

We **reach the conclusion that a similarity solution exists when $U_\infty(x)$ is a power law i.e. $U_\infty(x) = Cx^m$**. This is because we need β to be constant. The only mathematical way for β to stay constant is if in

$$\beta = \frac{g^2}{\nu U_\infty^2}\frac{dU_\infty}{dx} \tag{24}$$

$U_\infty(x) = Cx^m \rightarrow \frac{dU_\infty}{dx} = Cmx^{m-1}$ so, the dependence in x goes away. Noting that $g^2 \propto xU_\infty$.

Zero Pressure Gradient (m=0)

So, what value of m should we take on? The simplest one and the one that will be considered is $m = 0$. In this case, that means that $U_\infty = constant$. Recall that from Equation (16), this would also mean that $\frac{dP}{dx} = 0$.

$$U_\infty \frac{\partial U_\infty}{\partial x} = -\frac{1}{\rho}\frac{dP}{dx} \tag{16}$$

This is the scenario for a flat-plate, zero angle of attack scenario that will be used in the validation.

With $m = 0$, that gives us $\beta = 0$ and $\alpha = 1$ which reduces Equation (22) to:

$$f''' + ff'' = 0 \tag{35}$$

This is a non-linear ODE… there is no analytical solutions but there are numerical solutions that can be found readily online and in textbooks. I will not going into how those numerical solutions are obtained but they were done by Blasius using a shooting method (which is really just a try a guess for BC and see how close you are to your solution than update) and today can be done easily using your coding language of choice.

Validation

Approach

In order to validate the solution for a zero-pressure gradient, we can make a couple different comparisons for our laminar flat plate solution (recall that this corresponds to the m=0 solution). Specifically, the points of comparison will be the boundary layer profile and the skin friction coefficient.

Boundary-Layer Profile

The numerical solutions of $f'(\eta)$ (for which tables can be found online with f' values for given η) with that of the solution in the boundary layer from the SIMPLE solver.

Note what f' and η are (so that we can grab it from the solution).

$$\psi(x,y) = g(x)f(\eta) \tag{17}$$

$$\frac{\partial \psi}{\partial y} = g(x)\frac{\partial f}{\partial \eta}\frac{\partial \eta}{\partial y} = g(x)f'\left(\frac{U_\infty}{g(x)}\right) = U_\infty f' \tag{36}$$

$$\frac{\partial \psi}{\partial y} = u \tag{37}$$

Combining and rearranging Equation (36) and (37) we get:

$$f' = \frac{u}{U_\infty} \tag{38}$$

(we can easily get this from our solution field of the solver)

For η recall that it was defined as follows:

$$g^2 = U_\infty(2\alpha - \beta)\nu x \tag{29}$$

$$\eta = \frac{y}{\frac{g(x)}{U_\infty}} = \frac{yU_\infty}{\sqrt{U_\infty(2\alpha-\beta)\nu x}} \tag{39}$$

For the zero pressure gradient condition we are considering with $m = 0$ (and hence $\alpha = 1$ and $\beta = 0$) we end up with:

$$\eta = \frac{yU_\infty}{\sqrt{2U_\infty \nu x}} \tag{40}$$

(we can also easily get this from our solution field of the solver)

It is important to note that $x = 0$ is the leading edge as was used to determine the constant of integration in Equation (28). So we need to specify and use x accordingly.

So, we can plot the tabular values for η v.s. f' and compare it to those of the solution at some fixed x-location in our flow field.

Skin-Friction Coefficient

We can also validate against the skin friction coefficient along the plate (in the x-direction). The equation for the shear stress at the wall is as follows where we only consider the velocity-gradient in y since it is a no-slip wall so there is no variation in x.

$$\tau_w = \mu \frac{\partial u}{\partial y}_{wall} = \mu \left[\frac{\partial u}{\partial \eta}\frac{\partial \eta}{\partial y}\right]_{wall} \tag{41}$$

$$\frac{\partial u}{\partial \eta} = U_\infty f''(0) \tag{42}$$

$$\frac{\partial \eta}{\partial y} = \sqrt{\frac{U_\infty}{2\nu x}} \tag{43}$$

$$\tau_w = \mu U_\infty f''(0)\sqrt{\frac{U_\infty}{2\nu x}} \tag{44}$$

For the skin friction coefficient we simply divide this by the dynamic pressure.

$$C_f = \frac{\tau_w}{\frac{1}{2}\rho U_\infty^2} = 2f''(0)\sqrt{\frac{\nu}{2U_\infty x}} \tag{45}$$

Now this is skin friction is that derived from the tabulated values. How do we get the skin friction coefficient values from our solution? Just directly calculate the velocity gradient and use it in Equation (41) as below.

$$\tau_w = \mu \frac{\partial u}{\partial y}_{wall} = \mu\left(\frac{u_P - 0}{\Delta y}\right) \tag{46}$$

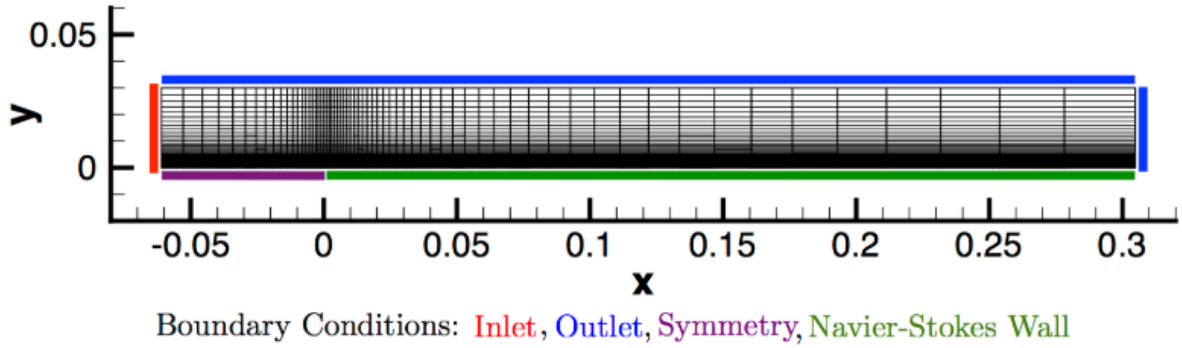
Notice the issue that comes up here. Your calculated wall shear stress will depend on the near wall grid refinement since a linear gradient is being computed when in reality the curve is quite non-linear… as we know from what BL profiles look like. We should make the mesh accordingly.

Mesh

As aforementioned, the refinement of the grid, specifically in the boundary layer is going to be important. We want enough cells to resolve the boundary layer profile and especially the first cell height which is imperative for calculating the skin friction.

The mesh (and boundary conditions) will follow that of this SU2 tutorial for a laminar flat plate where they also validate against the solution to the Blasius equation (this is actually where I got my inspiration from).

The mesh with the BCs from the SU2 case is shown below:



From: Laminar Flat Plate

The same boundary conditions will be imposed (except using inviscid wall instead of symmetry but those are the same BC in actuality).

Note that the upstream portion of the lower boundary uses a symmetry condition. If the entire bottom boundary were a no-slip wall, the inlet and wall boundary conditions would meet at the bottom-left corner, potentially introducing numerical complications depending on their implementation.

More importantly, the flow decelerates as it approaches the LE stagnation point, producing a rise in static pressure. Because the flow is subsonic, pressure disturbances can propagate upstream, allowing the approaching flow to respond to the airfoil's presence.

If the inlet were placed too close to the LE, its prescribed conditions could interfere with the developing pressure and velocity fields, potentially causing unphysical flow behavior or convergence difficulties. By extending the domain upstream and using a symmetry boundary, we provide a region where the flow can adjust before reaching the plate.

I will also be performing refinement near the leading edge of the flat plate as they have done and is good practice as lot of change occurs at that point in the x- and y- directions.

In their tutorial they say,

> In any simulation of viscous flow, it is important to capture the behavior of the boundary layer. Doing so requires an appropriate level of grid refinement near the wall. In this mesh, the vertical spacing is such that approximately 30 grid nodes lie within the boundary layer, which is typical for laminar flows of this nature.

From: Laminar Flat Plate

We will be aiming for 50 cells with a growth rate of 1.1 within the boundary layer. Outside of the boundary layer we will keep the grid visually fine enough and maintain a good enough growth rate so that transition from BL to freestream is not abrupt. Note that growth rates are pretty arbitrarily chosen but they should not exceed 1.2 which comes from general good practice (likely stemming from some validation research somewhere…)

To do this, we need to be able to determine the BL height (or at least approximate it). We can again use the Blasisus table of values. We know that $f' = \frac{u}{U_\infty}$. $f'(5)$ is close to 0.99 so we will use $\eta = 5$ to determine the approximate BL height where: (η is different here as the table I referenced used a different α in their formulation).

$$\eta = \frac{y}{\sqrt{\frac{\nu x}{U_\infty}}} \tag{47}$$

The flat plate will extend from 0 to 0.2 in my mesh so let's use $x = 0.1\ m$ and solve for y. As for the freestream conditions they will be $U_\infty = 34.03\ m/s$ and $\nu = 1.4607e\text{-}5\ \frac{m^2}{s}$. Therefore,

$$y = \eta\sqrt{\nu x U_\infty} = 5\sqrt{\frac{\left(1.4607e\text{-}5\ \frac{m^2}{s}\right)(0.1\ m)}{34.03\ \frac{m}{s}}} = 0.0010363\ m \tag{48}$$

So, we will want the aforementioned 50 cells in the BL to be between 0 m and 1.0363e-3 m.

This is more for my reference, so I am not going into the nitty gritty but the important part is resolving the BL fine enough and imposing the correct boundary conditions….

No conclusion or last remarks. This is the end… (I sleep now)